

Computational Finance — Lecture 4

Monte Carlo Simulation for Option Pricing

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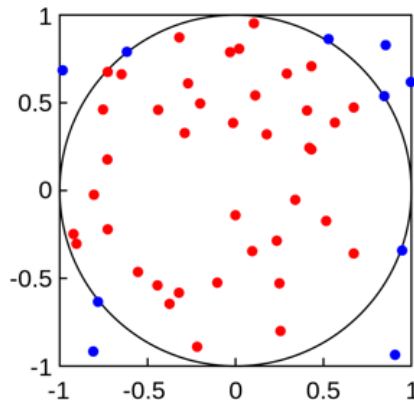
TU Delft

Lecture 4 - Monte Carlo Simulation for Option Pricing

- Monte Carlo Integration methods
- Monte Carlo simulation for stochastic differential equation
- Variance reduction techniques
 - Antithetic Variates
 - Control Variates
 - Quasi-Monte Carlo
 - Importance sampling
- A simple numerical experiment

Monte Carlo integration: background

- Monte Carlo integration is a technique for numerical integration using probability theory.
- Example:
 - The area of the circle (π) can be estimated by the ratio of the number of simulation points inside the circle to the total number of simulation points, yielding an approximation for the circle's area π .



https://en.wikipedia.org/wiki/Monte_Carlo_integration

Notions of convergence for random variables (1/2)

Let $\{X_n\}$ be a sequence of random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- **Almost sure convergence:** Let X be a random variable defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

$$X_n \xrightarrow{a.s.} X$$

means

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\right\}\right) = 1.$$

- **Convergence in probability:** Let X be a random variable defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

$$X_n \xrightarrow{p} X$$

means that for every $\epsilon > 0$,

$$\mathbb{P}(|X_n - X| > \epsilon) \rightarrow 0.$$

Notions of convergence for random variables (2/2)

- **Convergence in distribution:** Let X_n and X be random variables with cumulative distribution functions

$$F_{X_n}(x) = \mathbb{P}(X_n \leq x), \quad F_X(x) = \mathbb{P}(X \leq x).$$

Convergence in distribution

$$X_n \xrightarrow{d} X$$

means the distribution functions converge:

$$F_{X_n}(x) \rightarrow F_X(x)$$

for all x where F_X is continuous.

- **Function convergence:** The convergence above is **pointwise convergence of functions**: for every such x ,

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x).$$

Intuition: different types of convergence

- **Almost sure convergence** $X_n \xrightarrow{a.s.} X$

- For almost every outcome ω , the sequence $X_n(\omega) \rightarrow X(\omega)$.
- Interpretation: sample paths converge except on a set of probability zero.

- **Convergence in probability** $X_n \xrightarrow{p} X$

- For every $\varepsilon > 0$

$$\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0.$$

- Interpretation: large deviations from X become increasingly unlikely.

- **Convergence in distribution** $X_n \xrightarrow{d} X$

- The distribution functions converge:

$$F_{X_n}(x) \rightarrow F_X(x).$$

- Interpretation: the histograms / distributions converge.

Strength ordering of convergence

Strength hierarchy

The common notions of convergence satisfy the following implication chain:

$$X_n \xrightarrow{a.s.} X \Rightarrow X_n \xrightarrow{p} X \Rightarrow X_n \xrightarrow{d} X.$$

- **Almost sure convergence** is the strongest: the sample paths converge with probability one.
- **Convergence in probability** is weaker: the probability of large deviations goes to zero.
- **Convergence in distribution** is the weakest: only the distributions converge.

Important remark

The reverse implications do **not** generally hold. For example:

$$X_n \xrightarrow{d} X \not\Rightarrow X_n \xrightarrow{p} X.$$

Law of Large Numbers and Central Limit Theorem

Let $X_1, X_2, \dots$ be independent and identically distributed (i.i.d.) random variables with mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \text{Var}(X)$, and define the sample mean $\bar{X}_N = \frac{1}{N} \sum_{i=1}^N X_i$.

- **Strong Law of Large Numbers (SLLN)**

$$\bar{X}_N \xrightarrow{\text{a.s.}} \mu \quad \text{as } N \rightarrow \infty.$$

- **Weak Law of Large Numbers (WLLN)**

$$\bar{X}_N \xrightarrow{P} \mu \quad \text{as } N \rightarrow \infty.$$

- **Central Limit Theorem (CLT)**

$$\frac{\bar{X}_N - \mu}{\sigma/\sqrt{N}} \xrightarrow{d} N(0, 1).$$

Interpretation for Monte Carlo:

- LLN ensures the Monte Carlo estimator converges to the true value.
- CLT quantifies the estimation error and implies the standard error decays at rate $1/\sqrt{N}$.

Monte Carlo integration: 1D

- For one dimensional definite integral $I = \int_a^b f(x)dx$, we can rewrite I as

$$I = (b - a) \int_a^b \frac{f(x)}{b - a} dx = (b - a) \mathbb{E}[f(U)],$$

where U is a uniform random variable on $[a, b]$.

- By the strong law of large numbers, for an i.i.d. sequence of random variables $u_i \sim U[a, b]$,

$$\frac{1}{N} \sum_{i=1}^N f(u_i) \xrightarrow{a.s.} \mathbb{E}[f(U)] = \frac{I}{b - a}.$$

- Thus I can be estimated as

$$I \approx \frac{b - a}{N} \sum_{i=1}^N f(u_i).$$

Monte Carlo integration: multi-dimension

- In general, we can formulate the integration problem as

$$I = \int_{\Omega} f(\mathbf{x}) d\mathbf{x}$$

- Ω is the integration domain, a subset of $\mathbb{R}^d$;
 - $\mathbf{x} = [x_1, x_2, \dots, x_d]^T$ is a d -dimensional variable;
 - $f(\mathbf{x}) : \Omega \rightarrow \mathbb{R}$ is the integrand.
- Suppose we can factorize the integrand, i.e., $f(\mathbf{x}) = h(\mathbf{x})g(\mathbf{x})$, where $g(\mathbf{x})$ is the pdf of a random vector X .
- Then, it follows that

$$I = \int_{\Omega} h(\mathbf{x})g(\mathbf{x}) d\mathbf{x} = \mathbb{E}[h(X)].$$

- If $\mathbf{x}_i$ are i.i.d. samples from $g(\cdot)$, then by the strong law of large numbers,

$$\frac{1}{N} \sum_{i=1}^N h(\mathbf{x}_i) \xrightarrow{a.s.} \mathbb{E}[h(X)] = I.$$

Convergence of Monte Carlo integration

- Utilizing the fact that the sample variance

$$\text{Var}\left(\frac{1}{N} \sum_{i=1}^N h(x_i)\right) = \frac{1}{N} \text{Var}(h(X)) = \frac{\sigma^2}{N} \rightarrow 0,$$

the standard deviation of the Monte Carlo estimator decays at rate $\sqrt{N}$.

- By the Central Limit Theorem,

$$\frac{\bar{X}_N - I}{\sigma/\sqrt{N}} \xrightarrow{d} N(0, 1).$$

How to simulate stochastic integration?

- To answer this, we recall Itô's construction of the stochastic integral.
- Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $f(s, \omega)$ be a measurable process defined on $[0, \infty) \times \Omega$. The question Itô answered is: how can we define a stochastic integral

$$M_t(\omega) = \int_a^t f(s, \omega) dW_s(\omega), \quad a \leq t \leq b, \quad \omega \in \Omega.$$

where W_s is a Brownian motion, in such a way that the resulting process M_t is a martingale?

- This integral cannot be defined as a classical Riemann–Stieltjes integral.

Example why RS integral does not work (1/4)

- Recall the Riemann–Stieltjes integral:

$$\int_a^b f(t) dg(t) = \lim_{|\Delta_n| \rightarrow 0} \sum_{i=1}^n f(\tau_i)(g(t_i) - g(t_{i-1})),$$

where $\Delta_n = \{t_0, \dots, t_n\}$ is a partition of $[a, b]$ and $\tau_i \in [t_{i-1}, t_i]$.

- Consider the stochastic integral

$$I = \int_a^b W_t dW_t.$$

Using left endpoints,

$$L = \lim_{|\Delta_n| \rightarrow 0} \sum_{i=1}^n W_{t_{i-1}}(W_{t_i} - W_{t_{i-1}}).$$

- Introduce the corresponding right-endpoint sum

$$R = \lim_{|\Delta_n| \rightarrow 0} \sum_{i=1}^n W_{t_i}(W_{t_i} - W_{t_{i-1}}).$$

Example why RS integral does not work (2/4)

- First compute

$$\begin{aligned} L + R &= \lim_{|\Delta_n| \rightarrow 0} \sum_{i=1}^n (W_{t_{i-1}} + W_{t_i})(W_{t_i} - W_{t_{i-1}}) \\ &= \lim_{|\Delta_n| \rightarrow 0} \sum_{i=1}^n (W_{t_i}^2 - W_{t_{i-1}}^2) = W_b^2 - W_a^2. \end{aligned}$$

- Next compute

$$R - L = \lim_{|\Delta_n| \rightarrow 0} \sum_{i=1}^n (W_{t_i} - W_{t_{i-1}})^2 = b - a,$$

which follows from the quadratic variation of Brownian motion.

- Solving the system of equations for L and R gives

$$L = \frac{1}{2} (W_b^2 - W_a^2 - (b - a)), \quad R = \frac{1}{2} (W_b^2 - W_a^2 + (b - a)).$$

Example why RS integral does not work (3/4)

- Let $a = 0$ and $b = t$. Then

$$R_t = \frac{1}{2}(W_t^2 + t), \quad L_t = \frac{1}{2}(W_t^2 - t).$$

- Check the martingale property: Let $0 \leq s \leq t$ and write

$$W_t = W_s + (W_t - W_s).$$

Then

$$W_t^2 = W_s^2 + 2W_s(W_t - W_s) + (W_t - W_s)^2.$$

Since $W_t - W_s$ is independent of $\mathcal{F}_s$ and has mean 0 and variance $t - s$,

$$\mathbb{E}[W_t - W_s \mid \mathcal{F}_s] = 0, \quad \mathbb{E}[(W_t - W_s)^2 \mid \mathcal{F}_s] = t - s.$$

Example why RS integral does not work (4/4)

Therefore,

$$\mathbb{E}[W_t^2 \mid \mathcal{F}_s] = W_s^2 + 2W_s \mathbb{E}[W_t - W_s \mid \mathcal{F}_s] + \mathbb{E}[(W_t - W_s)^2 \mid \mathcal{F}_s] = W_s^2 + (t - s).$$

Hence,

$$\mathbb{E}[R_t \mid \mathcal{F}_s] = \frac{1}{2} \mathbb{E}[W_t^2 + t \mid \mathcal{F}_s] = \frac{1}{2} (W_s^2 + (t - s) + t) = \frac{1}{2} (W_s^2 + 2t - s) \neq R_s.$$

Conclusion: R_t is not a martingale.

On the other hand,

$$\mathbb{E}[L_t \mid \mathcal{F}_s] = \frac{1}{2} \mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = \frac{1}{2} (W_s^2 + (t - s) - t) = \frac{1}{2} (W_s^2 - s) = L_s.$$

Thus, L_t is a martingale.

Monte Carlo Simulation of SDE

- Suppose we are given the SDE

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t,$$

and we want to simulate X_T .

- The most intuitive and easy to implement discretization scheme is the **Euler–Maruyama scheme**.

- On a time grid

$$t_k = kh, \quad k = 0, 1, \dots, K, \quad Kh = T,$$

it is defined by

$$\hat{X}_{t_k} = \hat{X}_{t_{k-1}} + \mu(t_{k-1}, \hat{X}_{t_{k-1}}) h + \sigma(t_{k-1}, \hat{X}_{t_{k-1}}) \sqrt{h} Z_k.$$

where

- $\hat{X}_0 = X_0$ (same initial value),
- the simulation uses the grid $\{0, h, 2h, \dots, Kh = T\}$,
- Z_k are i.i.d. $N(0, 1)$.

Convergence of the Euler Scheme (1/2)

- There are two measures of the simulation error.
 - The simulation scheme has a strong order of convergence of $\beta > 0$, if

$$\mathbb{E}[|\hat{X}_{Kh} - X_T|] \leq c h^\beta,$$

for some constant c and all sufficiently small h .

- The simulation scheme has a weak order of convergence of $\beta > 0$, if

$$|\mathbb{E}[f(\hat{X}_{Kh})] - \mathbb{E}[f(X_T)]| \leq c h^\beta,$$

for some constant c , all sufficiently small h , and all $f \in C_p^{2\beta+2}$, where $C_p^{2\beta+2}$ denotes the class of smooth functions whose derivatives up to order $2\beta + 2$ have polynomial growth.

- The Euler scheme has a strong order of $\beta = \frac{1}{2}$ and a weak order of $\beta = 1$.

Convergence of the Euler Scheme (2/2)

- For the purpose of option pricing, we are more interested in the weak order of the convergence.
- However, in practice, the payoff functions, e.g., that of a European option, are not differentiable and thus not smooth.
- Thus, there is no theoretical guarantee that the Euler simulation scheme will have a weak order of convergence of $\beta = 1$.
- Simulation studies in (Broadie and Kaya 2006) suggest that the convergence rate of the Euler scheme for European option prices is $O(h^{1/3})$ or even slower.

Remark. The $1/3$ rate arises when balancing the discretization bias and the Monte Carlo variance under a fixed computational budget. Roughly,

$$\text{bias} \sim h, \quad \text{variance} \sim \frac{1}{N}, \quad \text{cost} \sim \frac{N}{h},$$

and optimizing the overall error yields a convergence rate of order $O(s^{-1/3})$, where s denotes the computational effort.

Milstein Scheme

- To achieve a strong order of convergence of $\beta = 1$, we can use the **Milstein scheme**.
- The **Milstein scheme** is given by

$$\begin{aligned}\hat{X}_{kh} &= \hat{X}_{t_{k-1}} + \mu\left(t_{k-1}, \hat{X}_{t_{k-1}}\right) h + \sigma\left(t_{k-1}, \hat{X}_{t_{k-1}}\right) \sqrt{h} Z_k \\ &\quad + \frac{1}{2} \sigma\left(t_{k-1}, \hat{X}_{t_{k-1}}\right) \frac{\partial \sigma}{\partial x}\left(t_{k-1}, \hat{X}_{t_{k-1}}\right) \left(Z_k^2 - 1\right) h,\end{aligned}$$

where the additional term involving the derivative of the diffusion coefficient accounts for the curvature of the diffusion and improves the strong convergence rate.

- **Convergence rates:**

strong order $\beta = 1$, weak order $\beta = 1$.

- However, it is not trivial to generalize this scheme to multi-dimensional SDEs.

Variance reduction techniques

- **Antithetic Variates:**

- The idea is that adding some simulation samples that are anticorrelated with the original random variable, which keeps the expected value unaltered and reduces the variance.

- **Control Variates:**

- using a known or estimated function to reduce the variance of the estimator. By subtracting a known or estimated function from the original estimator, we can reduce the variance of the estimator.

- **Quasi-Monte Carlo:**

- using deterministic sequences of points, such as low-discrepancy sequences, instead of random points for sampling. This can lead to a reduction in the variance of the estimator by improving the coverage of the sampling space.

- **Importance sampling:**

- instead of sampling from the original probability distribution of the problem, we sample from a different probability distribution that has the property of being easier to simulate or leads to a smaller variance of the estimator.
- Need to change the probability measures – Girsanov theorem.

Antithetic variates: uniform samples (1/3)

- To understand how the antithetic variate technique works, consider the more general case of approximating

$$I = \mathbb{E}[f(U)], \quad \text{where } U \sim U(0, 1)$$

for some function f .

- The standard Monte-Carlo estimate of I is

$$I_M = \frac{1}{M} \sum_{i=1}^M f(U_i) \quad \text{with i.i.d } U_i \sim U(0, 1)$$

$$\mathbb{E}[I_M] = I \quad \text{and} \quad \text{Var}(I_M) = \frac{1}{M} \text{Var}(f(U_i)).$$

- The estimator given by the antithetic variates method is

$$\hat{I}_M = \frac{1}{M} \sum_{i=1}^M \frac{f(U_i) + f(1 - U_i)}{2}$$

- Since $1 - U_i \sim U(0, 1)$,

Antithetic variates: uniform samples (2/3)

-

$$\begin{aligned}\text{Var}(\hat{I}_M) &= \frac{1}{M} \text{Var} \left(\frac{f(U_i) + f(1 - U_i)}{2} \right) \\ &= \frac{1}{M} \left[\frac{1}{4} \text{Var}(f(U_i)) + \frac{1}{4} \text{Var}(f(1 - U_i)) + \frac{1}{2} \text{Cov}(f(U_i), f(1 - U_i)) \right]\end{aligned}$$

- Note that

$$\text{Var}(f(U_i)) = \text{Var}(f(1 - U_i))$$

since both U_i and $1 - U_i$ follow the uniform distribution. Therefore

$$\text{Var}(\hat{I}_M) = \frac{1}{M} \left[\frac{1}{2} \text{Var}(f(U_i)) + \frac{1}{2} \text{Cov}(f(U_i), f(1 - U_i)) \right]$$

- The idea of the antithetic variates method is that if $\text{Cov}(f(U_i), f(1 - U_i)) < 0$, then $\text{Var}(\hat{I}_M) < \text{Var}(I_M)$ while $\mathbb{E}[\hat{I}_M] = \mathbb{E}[I_M] = I$.

Antithetic variates: uniform samples (3/3)

- **Lemma.** If f and g are both monotonic increasing functions or both monotonic decreasing functions then, for any random variable X ,

$$\text{Cov}(f(X), g(X)) > 0.$$

- Suppose $f(x)$ is a monotonic increasing (decreasing) function, then $-f(1 - x)$ is a monotonic increasing (decreasing) function as well.
- Therefore

$$\text{Cov}(f(U_i), -f(1 - U_i)) > 0,$$

which implies

$$\text{Cov}(f(U_i), f(1 - U_i)) < 0$$

and

$$\text{Var}(\hat{I}_M) < \text{Var}(I_M).$$

- In summary, if f is monotonic,

$$\text{Var}(\hat{I}_M) < \text{Var}(I_M).$$

Antithetic variates: normal samples

- The antithetic variates method is not restricted to uniform random variables.
- In the case of

$$I = \mathbb{E}[f(U)], \quad \text{where } U \sim \mathcal{N}(0, 1)$$

- The standard Monte Carlo estimate is

$$I_M = \frac{1}{M} \sum_{i=1}^M f(U_i), \quad \text{with i.i.d. } U_i \sim \mathcal{N}(0, 1)$$

- The estimator given by the antithetic variates method is

$$\hat{I}_M = \frac{1}{M} \sum_{i=1}^M \frac{f(U_i) + f(-U_i)}{2}$$

- Note that both U_i and $-U_i$ follow the standard Normal distribution. Therefore
 - $\mathbb{E}[\hat{I}_M] = \mathbb{E}[I_M] = I$
 - $\text{Var}(\hat{I}_M) < \text{Var}(I_M)$ if $\text{Cov}(f(U_i), f(-U_i)) < 0$ (which holds, e.g., if f is monotonic).

Antithetic variates: Multivariate case

- The antithetic variates idea extends readily to the case where f is a function of more than one random variable. For example, suppose we wish to approximate

$$I = \mathbb{E}[f(U, V, W)],$$

where U , V , and W are i.i.d. $\mathcal{N}(0, 1)$.

- The standard Monte Carlo estimate is

$$I_M = \frac{1}{M} \sum_{i=1}^M f(U_i, V_i, W_i),$$

with U_i , V_i , and W_i i.i.d. $\mathcal{N}(0, 1)$.

- And the antithetic version is

$$\hat{I}_M = \frac{1}{M} \sum_{i=1}^M \frac{f(U_i, V_i, W_i) + f(-U_i, -V_i, -W_i)}{2}.$$

- If f is monotonic in each of the arguments, $\text{Var}(\hat{I}_M) < \text{Var}(I_M)$.

Antithetic variates in option valuation (1/2)

- The application of the antithetic variates method is Monte Carlo estimation of path-dependent exotic options.
- Suppose that along a simulation path, we simulate a sequence of i.i.d. $\mathcal{N}(0, 1)$:

$$Z_0, Z_1, \dots, Z_{N-1}.$$

- The antithetic variates method simulates another path

$$-Z_0, -Z_1, \dots, -Z_{N-1}$$

and averages the payoffs from the two paths.

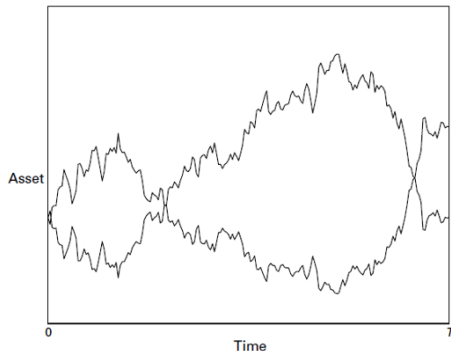


Figure: A pair of discrete asset paths computed using antithetic variates. The payoff from both paths is averaged to give a single sample.

Antithetic variates in option valuation (2/2)

The algorithm of the antithetic variates method for valuing an up-and-in call option is as follows:

```
for  $i = 1$  to  $M$ 
  for  $j = 0$  to  $N - 1$ 
    compute an  $N(0, 1)$  sample  $\xi_j$ 
    set  $S_{j+1} = S_j e^{(r - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\xi_j}$ 
    set  $\bar{S}_{j+1} = \bar{S}_j e^{(r - \frac{1}{2}\sigma^2)\Delta t - \sigma\sqrt{\Delta t}\xi_j}$ 
  end
  set  $S_i^{\max} = \max_{0 \leq j \leq N} S_j$ 
  set  $\bar{S}_i^{\max} = \max_{0 \leq j \leq N} \bar{S}_j$ 
  if  $S_i^{\max} > B$  set  $V_i = e^{-rT} \max(S_N - E, 0)$ , otherwise  $V_i = 0$ 
  if  $\bar{S}_i^{\max} > B$  set  $\bar{V}_i = e^{-rT} \max(\bar{S}_N - E, 0)$ , otherwise  $\bar{V}_i = 0$ 
  set  $\hat{V}_i = \frac{1}{2}(V_i + \bar{V}_i)$ 
end
set  $a_M = \frac{1}{M} \sum_{i=1}^M \hat{V}_i$ 
set  $b_M^2 = \frac{1}{M-1} \sum_{i=1}^M (\hat{V}_i - a_M)^2$ 
```

Control variates (1/2)

- The control variates method is a **variance reduction** technique used in Monte Carlo methods.
- Suppose that we want to estimate $\mathbb{E}[X]$ via simulation. In the meanwhile, we can easily simulate a correlated random variable Y , and moreover, $\mathbb{E}[Y]$ is known.
- Obviously we can simulate the random variable

$$Z = X + \mathbb{E}[Y] - Y$$

since $\mathbb{E}[Z] = \mathbb{E}[X]$.

- We want to choose Y such that

$$\text{Var}(Z) = \text{Var}(X - Y) < \text{Var}(X).$$

Control variates (2/2)

- We make use of the generalized expression: $Z_\theta = X + \theta(\mathbb{E}[Y] - Y)$
- It follows that $\text{Var}(Z_\theta) = \text{Var}(X) - 2\theta \text{Cov}(X, Y) + \theta^2 \text{Var}(Y)$
- Since

$$\text{Var}(Z_\theta) = \text{Var}(Y) \left(\theta - \frac{\text{Cov}(X, Y)}{\text{Var}(Y)} \right)^2 + \text{Var}(X) - \frac{\text{Cov}(X, Y)^2}{\text{Var}(Y)},$$

the θ that minimizes $\text{Var}(Z_\theta)$ is

$$\theta_{\min} = \frac{\text{Cov}(X, Y)}{\text{Var}(Y)}.$$

Correspondingly,

$$\text{Var}(Z_{\theta_{\min}}) = (1 - \rho_{XY}^2) \text{Var}(X),$$

where ρ_{XY} is the correlation between X and Y .

- On a general problem we may not know $\text{Cov}(X, Y)$ and $\text{Var}(Y)$, and hence cannot find $\theta_{\min}$. However, it is possible to estimate $\text{Cov}(X, Y)$ and $\text{Var}(Y)$ during a MC simulation.

Control variates in option valuation

- The control variate idea can be used on path-dependent options where there is no known analytical expression for the option value, but there is an expression for a similar option.
- The classic example is an *arithmetic average price Asian option*, where the average is taken over a pre-set collection of discrete times $\{t_i\}_{i=1}^n$. The payoff for the arithmetic average price Asian call option is $\max\left(\frac{1}{n} \sum_{i=1}^n S_{t_i} - K, 0\right)$, while the payoff of the geometric average price Asian option is $\max\left(\left(\prod_{i=1}^n S_{t_i}\right)^{1/n} - K, 0\right)$.
- Obviously the prices of the arithmetic and geometric average options are correlated, since their payoffs are correlated along every path $\{S_{t_i}\}_{i=1}^n$.
- If $\{t_i\}_{i=1}^n$ are equally spaced, there is an exact formula for the geometric average option. Hence we may use the geometric average option as a control variate when valuing the arithmetic version.

Quasi-Monte Carlo Simulation (1/3)

Quasi-Monte Carlo (QMC) replaces random samples with **low-discrepancy sequences** that fill the space more uniformly:

$$u_1, u_2, \dots, u_N \in [0, 1]^d.$$

Examples:

- Sobol sequence
- Halton sequence
- Faure sequence
- Lattice rules

Advantages:

- Better space coverage than random sampling
- Often faster convergence than standard Monte Carlo

Quasi-Monte Carlo Simulation (2/3)

- For Quasi-Monte Carlo, the Koksma–Hlawka inequality gives

$$\left| \frac{1}{N} \sum_{i=1}^N f(x_i) - \int f(x) dx \right| \leq V(f) D_N^*,$$

- $V(f)$ is the the Hardy–Krause variation, which measures the total magnitude of all mixed derivatives of the function across coordinate subsets, quantifying how rough the integrand is.
- D_N^* is the star discrepancy of a point set $P_N = \{x_1, \dots, x_N\} \subset [0, 1]^d$, defined as

$$D_N^*(P_N) = \sup_{t \in [0, 1]^d} \left| \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{x_i \in [0, t)} - \prod_{j=1}^d t_j \right|,$$

with $[0, t) = [0, t_1) \times \dots \times [0, t_d)$. It measures the maximum deviation between the empirical fraction of points in origin-anchored boxes and their volume.

Quasi-Monte Carlo Simulation (3/3)

- Low-discrepancy sequences (Sobol, Halton, lattice) yield:

$$D_N^* = \mathcal{O} \left(\frac{(\log N)^d}{N} \right).$$

- Thus, for sufficiently smooth f , QMC convergence rate is:

$$\mathcal{O} \left(\frac{(\log N)^d}{N} \right).$$

Monte Carlo (MC) vs Quasi-Monte Carlo (QMC)

- **Monte Carlo (MC):**
 - Random sampling.
 - Convergence rate: $\mathcal{O}(N^{-1/2})$.
 - Rooted in probability theory.
- **Quasi-Monte Carlo (QMC):**
 - Deterministic, low-discrepancy sequences.
 - Convergence rate: $\mathcal{O}((\log N)^d/N)$ for low-discrepancy sequences.
 - Based on discrepancy theory.
- Both methods use $\frac{1}{N} \sum_{i=1}^N f(x_i)$, but differ in how points x_i are chosen.

Comparison with Classical Quadrature

- **Quadrature rules:**

- A general quadrature rule approximates an integral by a weighted sum of function evaluations:

$$\int_{\Omega} f(x) dx \approx \sum_{i=1}^N w_i f(x_i),$$

where x_i are the quadrature points and w_i are the associated weights.

- Exact for polynomials up to a certain degree.
- Rapid convergence, e.g., $\mathcal{O}(N^{-k})$, but suffers from the curse of dimensionality.
- Effective in low dimensions (e.g., $d \leq 3$).

- **QMC vs Quadrature rules:**

- QMC scales better in higher dimensions.
- No exact polynomial integration, but minimizes "discrepancy" (uniformity of points).

Lattice Rules as a QMC Family

- Lattice rules are a subset of QMC methods.
- They generate integration points via a structured, modular arithmetic pattern:

$$x_n = \left\{ \frac{nz}{N} \right\} \pmod{1},$$

where z is a generating vector.

- Particularly effective for periodic and smooth integrands.
- Often combined with random shifts to estimate error.

Transforming to Non-Unit Domains

- QMC sequences are typically defined on $[0, 1]^d$.
- For an arbitrary domain $[a, b]^d$, apply a linear transformation:

$$x_i^{(\text{new})} = a + (b - a) \cdot x_i.$$

- For integrals over $\mathbb{R}^d$ (e.g., Gaussian integrals), use transformations like:

$$x_i = \Phi^{-1}(u_i), \quad u_i \in [0, 1],$$

where Φ^{-1} is the inverse CDF of the target distribution.

- **The idea:**

- certain values of the input random variables in a simulation have more impact on the parameter being estimated than others.
- If these “important” values are emphasized by sampling more frequently, then the estimator variance can be reduced.
- Hence, the basic methodology in importance sampling is to choose a distribution which “encourages” the important values.
- This use of “biased” distributions will result in a biased estimator if it is applied directly in the simulation.
- However, the simulation outputs are weighted to correct for the use of the biased distribution, and this ensures that the new importance sampling estimator is unbiased.
- The weight is given by the **likelihood ratio**, that is, the Radon–Nikodym derivative of the true underlying distribution with respect to the proposal (biased) distribution.

Importance Sampling (2/4)

Suppose we want to estimate

$$\mathbb{E}_p[f(X)] = \int f(x)p(x) dx.$$

Instead of sampling X from $p(x)$, choose another distribution $q(x)$ (the **proposal distribution**) and write

$$\mathbb{E}_p[f(X)] = \mathbb{E}_q\left[f(X)\frac{p(X)}{q(X)}\right].$$

- Samples are drawn from $q(x)$ instead of $p(x)$.
- The factor $\frac{p(X)}{q(X)}$ corrects the bias.
- A good proposal distribution places more probability mass where $f(x)p(x)$ is large.

Importance Sampling (3/4)

- The fundamental issue is the choice of the **proposal (biased) distribution** that emphasizes the important regions of the input variables.
- Designing a good proposal distribution is often considered the *art* of importance sampling.
- A well-chosen distribution can lead to **dramatic reductions in simulation variance and run time**.
- Conversely, a poorly chosen distribution may lead to **worse performance than standard Monte Carlo**.

Importance Sampling (4/4)

- Importance sampling evaluates an expectation under one distribution by sampling under another:

$$\mathbb{E}_p[f(X)] = \mathbb{E}_q\left[f(X)\frac{p(X)}{q(X)}\right].$$

Here q is chosen to make the simulation more efficient. There is no requirement that q comes from a numeraire.

- Change of numeraire is a special type of change of measure defined by choosing a tradable asset as the numeraire. If N_t is the numeraire and M_t is the bank account, the associated probability measure Q^N is defined through the Radon–Nikodym derivative

$$\left.\frac{dQ^N}{dQ}\right|_{\mathcal{F}_T} = \frac{N_T/M_T}{N_0/M_0}.$$

Under the measure Q^N , the price of any asset expressed in units of the numeraire, S_t/N_t , is a martingale.

Importance Sampling: Drift Shifting (1/3)

- Importance sampling via *drift shifting* modifies the sampling distribution so that paths leading to large payoffs are sampled more frequently. Suppose

$$V = \mathbb{E}[f(X)], \quad X \sim \mathcal{N}(0, I_d).$$

Instead of sampling X directly, we introduce a drift shift θ and simulate

$$X = Z + \theta, \quad Z \sim \mathcal{N}(0, I_d).$$

- To preserve unbiasedness, the likelihood ratio (Radon–Nikodym derivative) is applied:

$$V = \mathbb{E}\left[f(Z + \theta) \exp\left(-\theta^\top Z - \frac{1}{2}\|\theta\|^2\right)\right].$$

- By choosing θ so that the shifted distribution places more probability mass in regions where the payoff f is large (e.g., in-the-money scenarios for options), the variance of the Monte Carlo estimator can be significantly reduced.

Importance Sampling: Drift Shifting (2/3)

How to choose the optimal drift shift θ ?

- We estimate

$$V = \mathbb{E}[f(X)], \quad X \sim \mathcal{N}(0, I_d),$$

using importance sampling with a drift shift θ . Let $Z \sim \mathcal{N}(0, I_d)$ and $X = Z + \theta$. The estimator becomes

$$\hat{V} = f(Z + \theta) \exp\left(-\theta^\top Z - \frac{1}{2}\|\theta\|^2\right).$$

- The optimal shift θ is chosen by minimizing the variance of the estimator. Equivalently, we minimize the second moment

$$M(\theta) = \mathbb{E}\left[f(Z + \theta)^2 \exp\left(-2\theta^\top Z - \|\theta\|^2\right)\right].$$

- The optimal parameter satisfies the first-order condition

$$\nabla_\theta M(\theta^*) = 0,$$

which determines the variance-minimizing importance sampling drift.

Importance Sampling: Drift Shifting (3/3)

Example: estimate $C_0 = e^{-rT} \mathbb{E}^Q[(S_T - K)^+]$ under the Black–Scholes model.

- Under Q ,

$$S_T = S_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z\right), \quad Z \sim N(0, 1).$$

- Instead of sampling $Z \sim N(0, 1)$, we simulate $Z^* \sim N(\mu, 1)$.
- Correct the bias using the likelihood ratio

$$L(Z^*) = \exp\left(-\mu Z^* + \frac{1}{2}\mu^2\right).$$

- Proof: Let f and g be the density functions of $N(0, 1)$ and $N(\mu, 1)$, respectively:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad g(z) = \frac{1}{\sqrt{2\pi}} e^{-(z-\mu)^2/2}$$

Then,

$$L(z) = \frac{f(z)}{g(z)} = \exp\left(-\frac{z^2}{2} + \frac{(z-\mu)^2}{2}\right) = \exp\left(-\mu z + \frac{1}{2}\mu^2\right).$$

- The estimator becomes $\hat{C}_N = \frac{1}{N} \sum_{i=1}^N e^{-rT} (S_T^{(i)} - K)^+ L(Z_i^*)$. Choose μ shifts probability mass toward the payoff region $S_T > K$, reducing simulation variance.

Change of Numeraire (1/2)

- A **numeraire** is the price process of a strictly positive tradable asset that can be used as a unit of value.
- Mathematically, this means that the numeraire process N_t is a strictly positive adapted stochastic process on the probability space $(\Omega, \mathcal{F}, P)$.
- Under the risk-neutral measure Q associated with the bank account numeraire M_t , the discounted price of any asset S_t satisfies $\frac{S_t}{M_t}$ is a martingale.

Change of Numeraire (2/2)

- Suppose we want to use N_t as the numeraire instead of M_t . We need to find the new risk-neutral measure Q^N such that under Q^N , the discounted price of any asset S_t with respect to N_t is a martingale, i.e.,

$$\frac{S_t}{N_t} \text{ is a } Q^N\text{-martingale.}$$

- Since M_t is the numeraire under Q , we have $\frac{N_0}{M_0} = \mathbb{E}^Q \left[\frac{N_T}{M_T} \right]$. Then

$$Z_t^{Q \rightarrow N} := \frac{M_0 N_t}{N_0 M_t}.$$

is $\mathcal{F}_T$ measurable and satisfies $\mathbb{E}^Q[Z_T^{Q \rightarrow N}] = 1$.

- Thus, $Z_t^{Q \rightarrow N}$ is the density process for changing from Q to the measure corresponding to using N_t as the numeraire.
- The reverse density process is $Z_t^{N \rightarrow Q} = \frac{1}{Z_t^{Q \rightarrow N}} = \frac{N_0 M_t}{M_0 N_t}$.

Example: Using Stock as Numeraire (1/4)

- Recall the Black–Scholes model under the risk-neutral measure Q :

$$dS_t = rS_t dt + \sigma S_t dW_t^Q.$$

- Consider a European call with maturity T :

$$C_0 = e^{-rT} \mathbb{E}^Q[(S_T - K)^+] = e^{-rT} \mathbb{E}^Q[S_T 1_{\{S_T > K\}}] - e^{-rT} K \mathbb{E}^Q[1_{\{S_T > K\}}].$$

- Instead, we use S_t as the numeraire and denote the corresponding measure by Q^S .
 - By change of measure, we obtain

$$\mathbb{E}^Q[S_T 1_{\{S_T > K\}}] = \mathbb{E}^{Q^S} \left[S_T 1_{\{S_T > K\}} \frac{dQ}{dQ^S} \Big|_{\mathcal{F}_T} \right] = \mathbb{E}^{Q^S} \left[S_T 1_{\{S_T > K\}} Z_T^{S \rightarrow Q} \right].$$

- Substituting $Z_T^{S \rightarrow Q}$ and inserting $M_T = e^{rT} M_0$ give

$$= \mathbb{E}^{Q^S} \left[S_T 1_{\{S_T > K\}} \frac{M_T S_0}{S_T M_0} \right] = \frac{S_0}{M_0} \mathbb{E}^{Q^S} [M_T 1_{\{S_T > K\}}] = e^{rT} S_0 \mathbb{E}^{Q^S} [1_{\{S_T > K\}}].$$

- Thus, computing $\mathbb{E}^Q[S_T 1_{\{S_T > K\}}]$ reduces to computing the probability $\mathbb{E}^{Q^S}[1_{\{S_T > K\}}]$.

Example: Using Stock as Numeraire (2/4)

- From the change-of-numeraire formula, the Radon–Nikodym derivative of the stock-numeraire measure Q^S with respect to Q is

$$Z_t^{Q \rightarrow S} = \frac{dQ^S}{dQ} \Big|_{\mathcal{F}_t} = \frac{M_0 S_t}{S_0 M_t} = e^{-rt} \frac{S_t}{S_0}.$$

- Under the Black–Scholes model

$$dS_t = rS_t dt + \sigma S_t dW_t^Q,$$

we have

$$d(e^{-rt} S_t) = e^{-rt} dS_t - re^{-rt} S_t dt = e^{-rt} \sigma S_t dW_t^Q.$$

Therefore

$$dZ_t^{Q \rightarrow S} = \frac{1}{S_0} d(e^{-rt} S_t) = \sigma \frac{e^{-rt} S_t}{S_0} dW_t^Q = \sigma Z_t^{Q \rightarrow S} dW_t^Q.$$

Example: Using Stock as Numeraire (3/4)

- Since

$$dZ_t^{Q \rightarrow S} = \sigma Z_t^{Q \rightarrow S} dW_t^Q, \quad Z_0^{Q \rightarrow S} = 1,$$

the solution is the stochastic exponential

$$Z_t^{Q \rightarrow S} = \exp\left(\int_0^t \sigma dW_s^Q - \frac{1}{2} \int_0^t \sigma^2 ds\right) = \exp\left(\sigma W_t^Q - \frac{1}{2} \sigma^2 t\right).$$

- Comparing this with Girsanov's theorem, we have

$$Z_t^{Q \rightarrow S} = \exp\left(\int_0^t Y_s dW_s^Q - \frac{1}{2} \int_0^t Y_s^2 ds\right)$$

with $Y_s \equiv \sigma$. Therefore, under the measure Q^S defined by

$$\left. \frac{dQ^S}{dQ} \right|_{\mathcal{F}_t} = Z_t^{Q \rightarrow S},$$

the process

$$W_t^S = W_t^Q - \int_0^t Y_s ds = W_t^Q - \sigma t$$

Example: Using Stock as Numeraire (4/4)

Under the stock-numeraire measure Q^S we derived that

$$dS_t = (r + \sigma^2)S_t dt + \sigma S_t dW_t^S.$$

Applying Itô's lemma to $\log S_t$,

$$d \log S_t = \left(r + \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t^S.$$

Hence,

$$\log S_T = \log S_0 + \left(r + \frac{1}{2} \sigma^2 \right) T + \sigma W_T^S.$$

Therefore,

$$Q^S(S_T > K) = Q^S(\log S_T > \log K) = N(d_1),$$

where

$$d_1 = \frac{\log(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

Summary

Run some simulations to compare the performance of the MC methods for valuing an European option.

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